Name: Akshita Agarwal
Registration Number: 24BDS0125

Lab Assignment – 02

Generate Lexical Analyzer using Lex Program

Question 1: A data entry system needs to track the number of alphabets and digits entered by a user in a form field for statistical analysis. Write a Lex program that reads an input string and counts number of alphabets and digits.

Program:

Open | lex1.l ~/24BDS0125 | Save

```
%{
#include<stdio.h>
int a=0;
int d=0;
%}
ALP [A-Za-z]
DIGIT [0-9]

%%
{ALP}   {a++;}
{DIGIT} {d++;}
.       {printf(" %s is Invalid",yytext);}

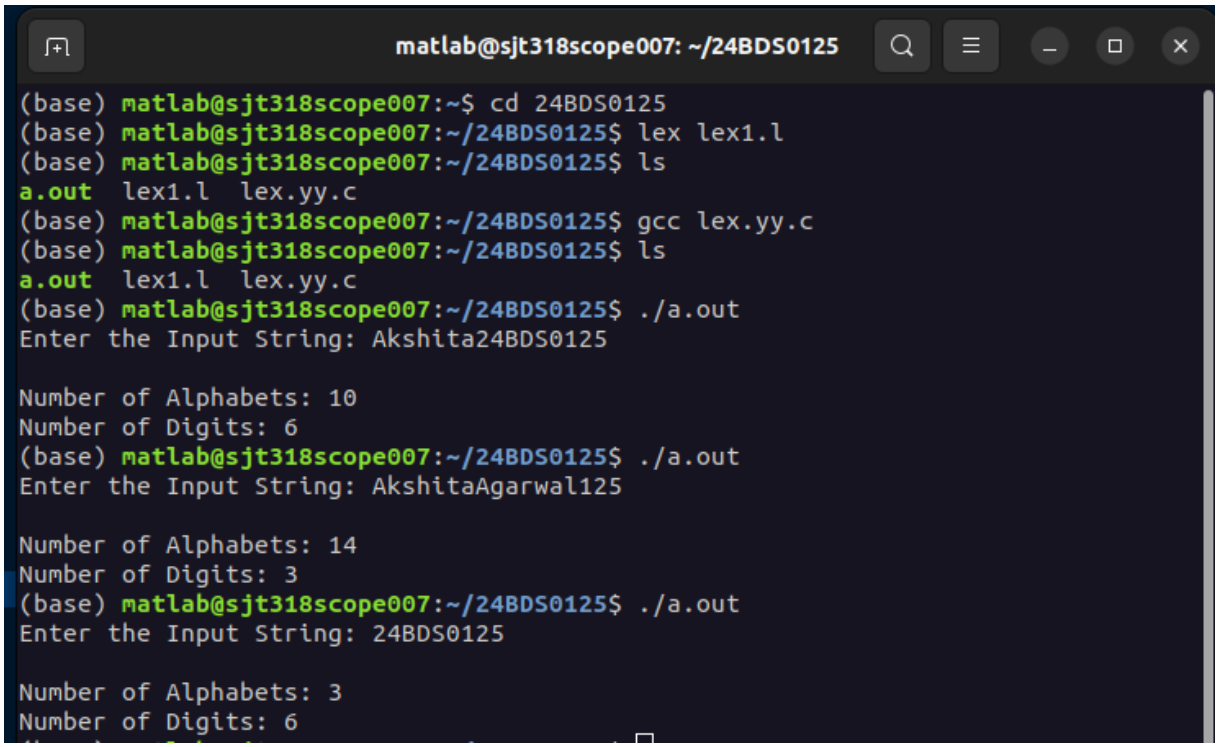

%%
int main()
{
        printf("Enter the Input String: ");
        yylex();
        printf("Number of Alphabets: %d\n",a);
        printf("Number of Digits: %d\n",d);
        return 0;
}

int yywrap()
{
        return 1;
}

```

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Input/Output Screenshots:



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Question 2: A typing tutor software is designed to monitor a student's typing performance during practice sessions. As part of the evaluation, it needs to keep track of how many vowels and consonants the student has typed. Write a Lex program that reads an input text and counts the number of vowels (both uppercase and lowercase) and the number of consonants (both uppercase and lowercase).

Program:

Open | lex2.l ~/24BDS0125 | Save

lex1.l | lex2.l

```
%{
#include<stdio.h>
int v=0;
int c=0;
%}
VOW a|e|i|o|u|A|E|I|O|U
CON b|c|d|f|g|h|j|k|l|m|n|p|q|r|s|t|v|w|x|y|z|B|C|D|F|G|H|J|K|L|M|N|P|Q|R|S|T|V|W|X|Y|Z

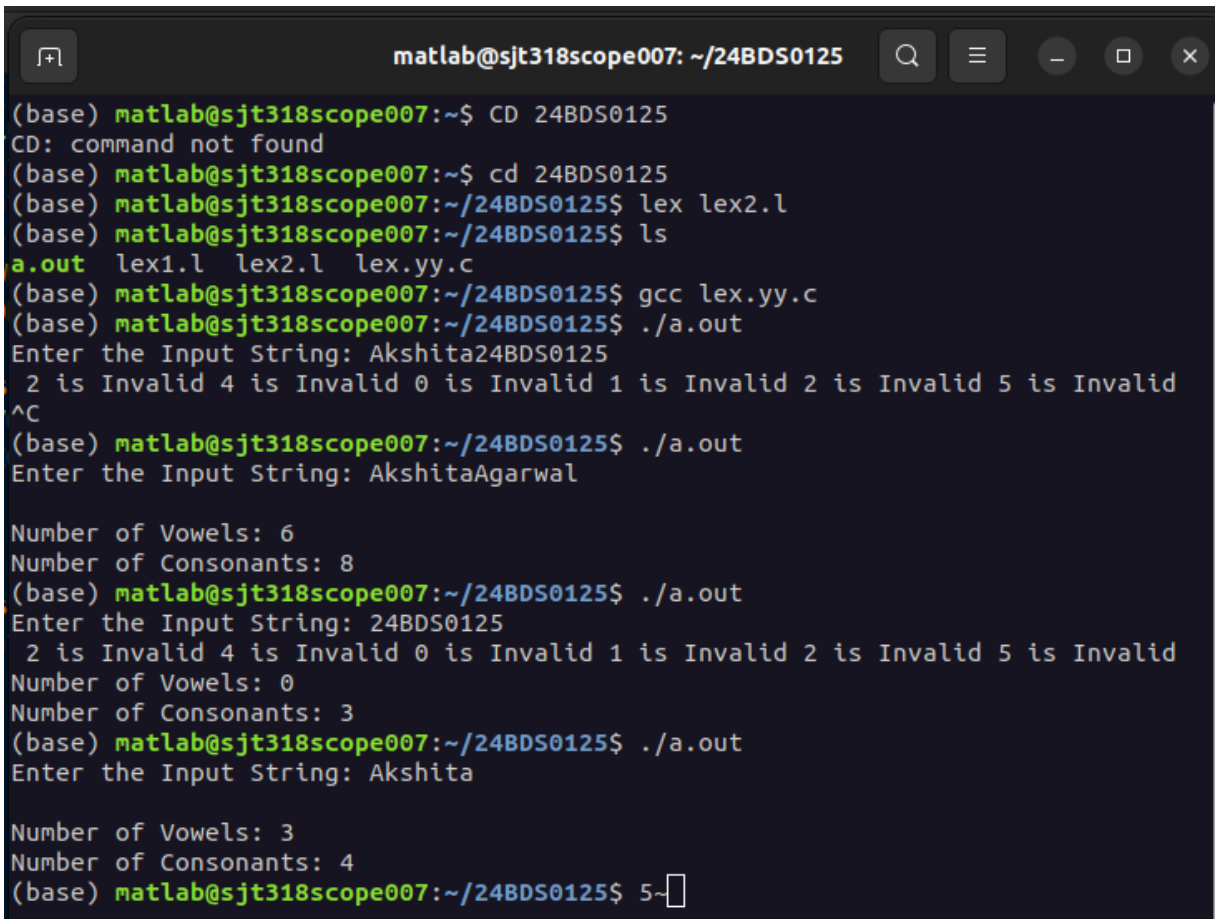
%%
{VOW}   {v++;}
{CON} {c++;}
.       {printf(" %s is Invalid",yytext);}


%%
int main()
{
        printf("Enter the Input String: ");
        yylex();
        printf("Number of Vowels: %d\n",v);
        printf("Number of Consonants: %d\n",c);
        return 0;
}

int yywrap()
{
        return 1;
}
```

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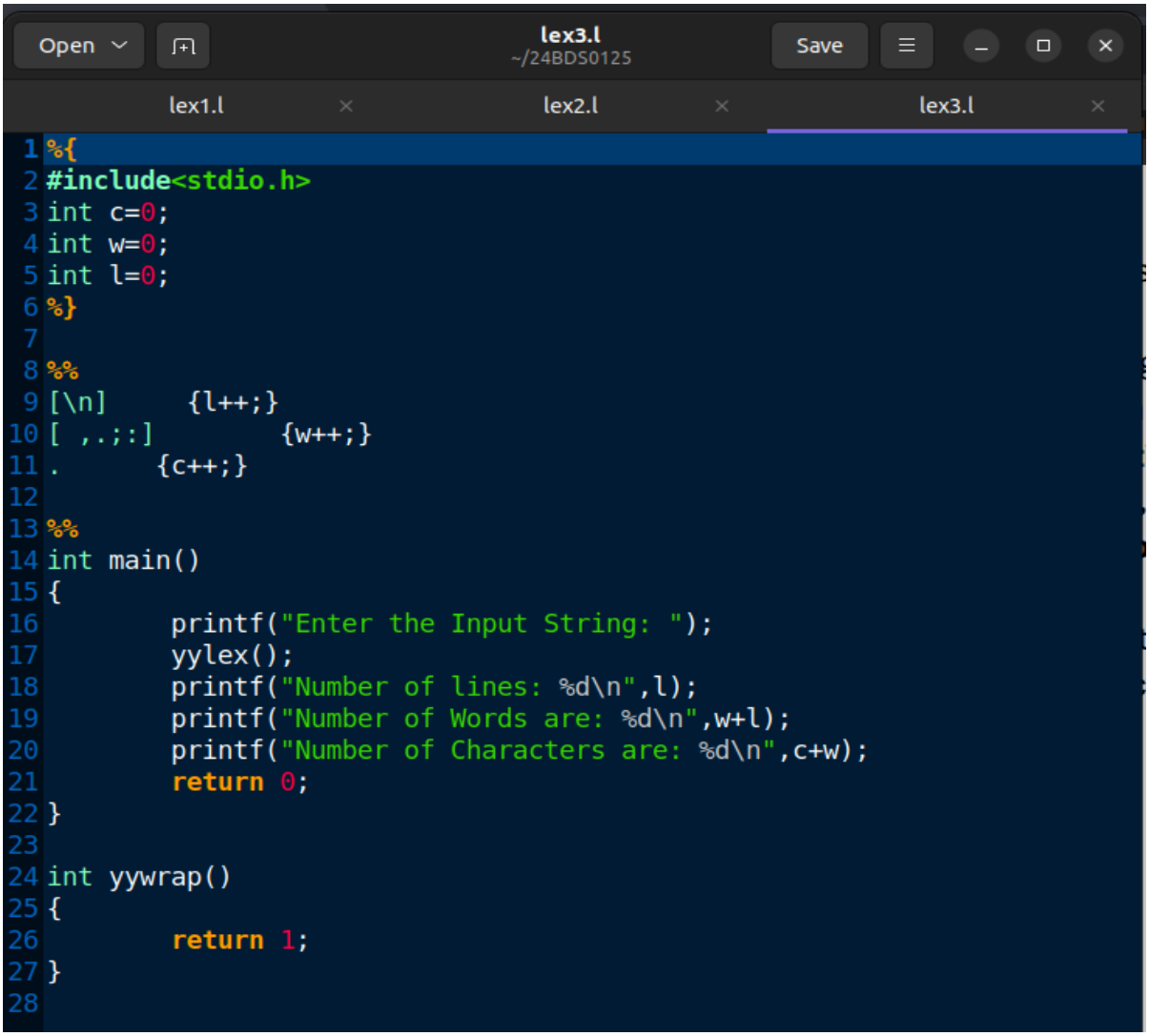
Input/Output Screenshots:



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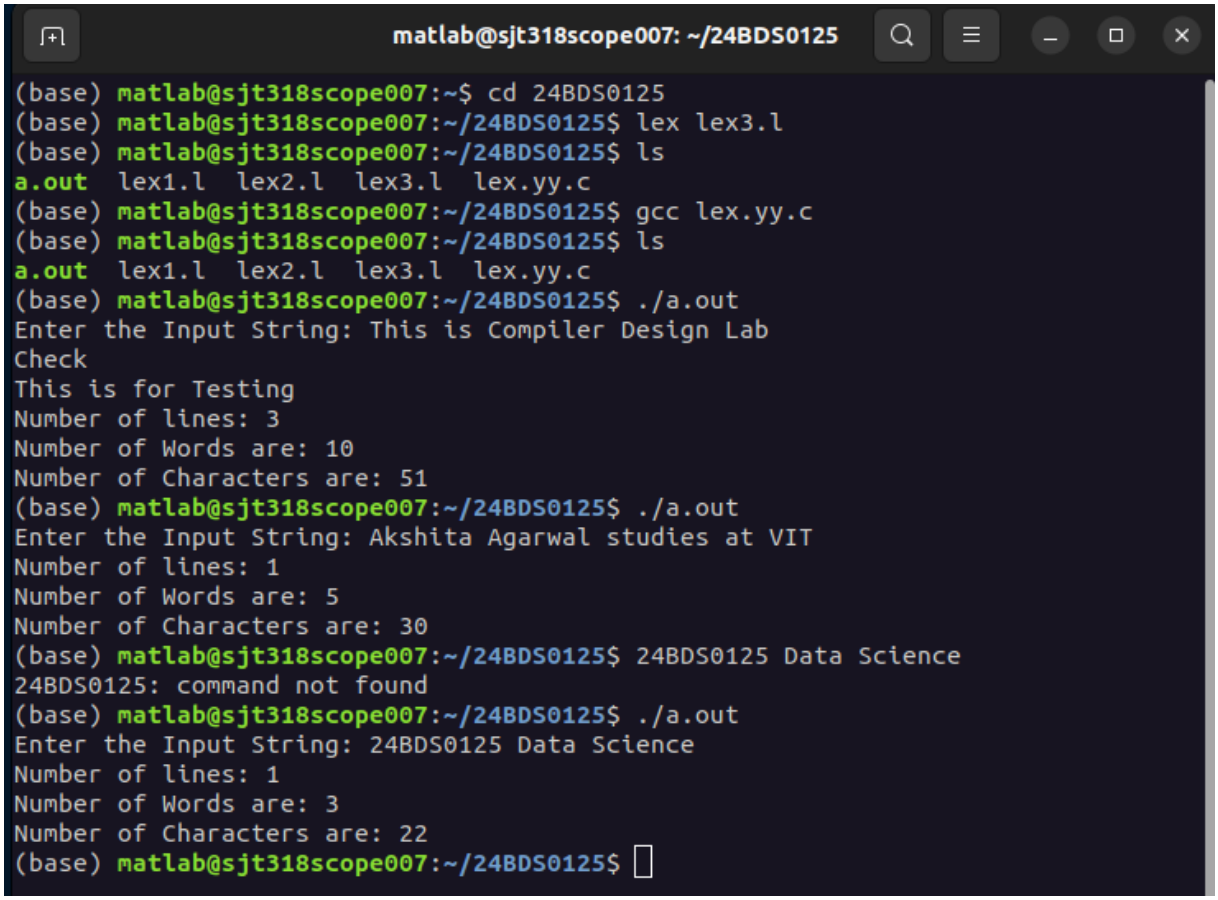
Question 3: A text analysis tool needs to provide quick statistics about any given document or text input. As part of its functionality, it should count the total number of lines, words, and characters entered by the user. Write a Lex program that reads an input stream and counts the number of lines, words and characters.

Program:



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Input/Output Screenshots:

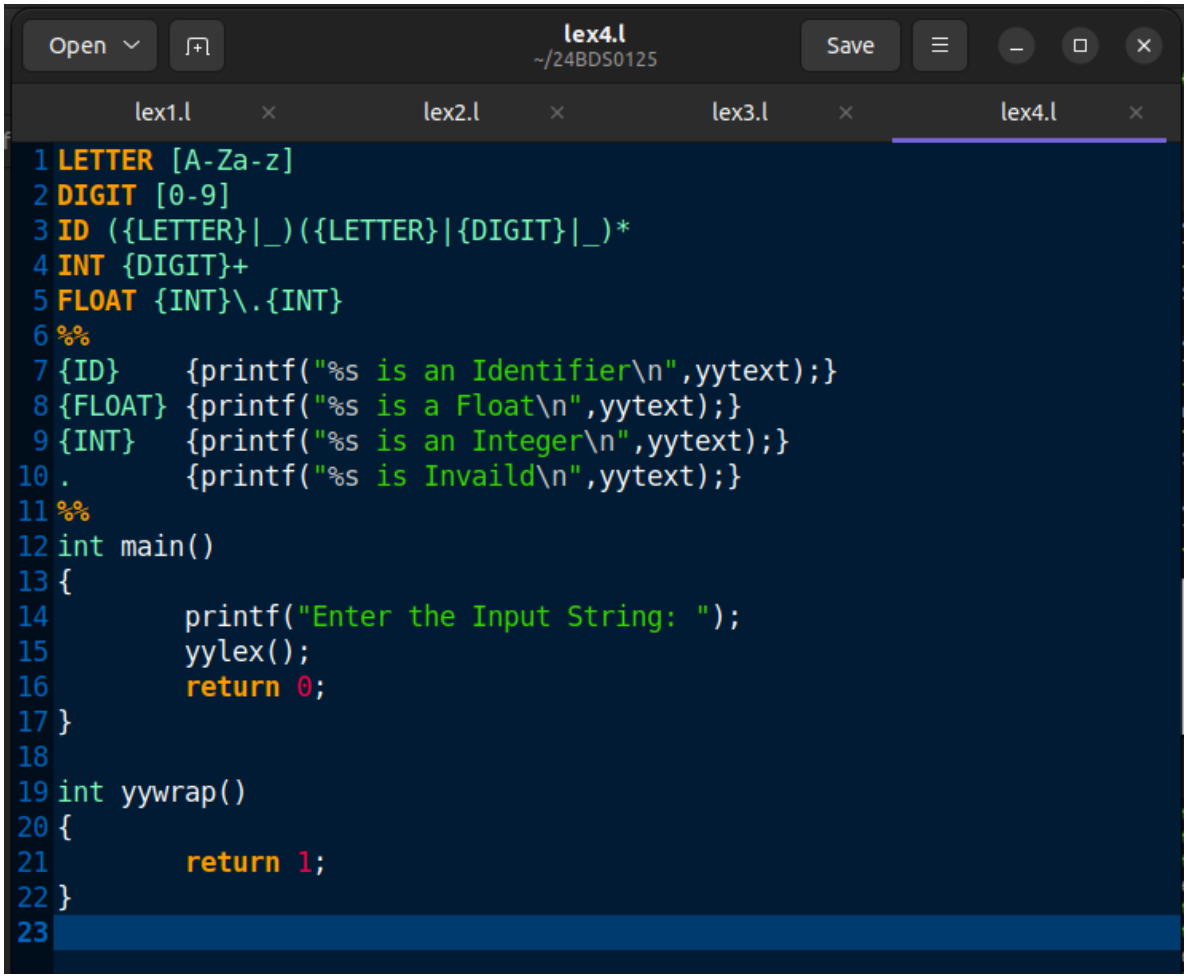


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Question 4: In the lexical analysis phase of a compiler's front-end, the system must verify whether an input string is a valid variable name (identifier) or a valid number (constant) according to programming language rules. Write a Lex program that reads an input string and determines whether it is:

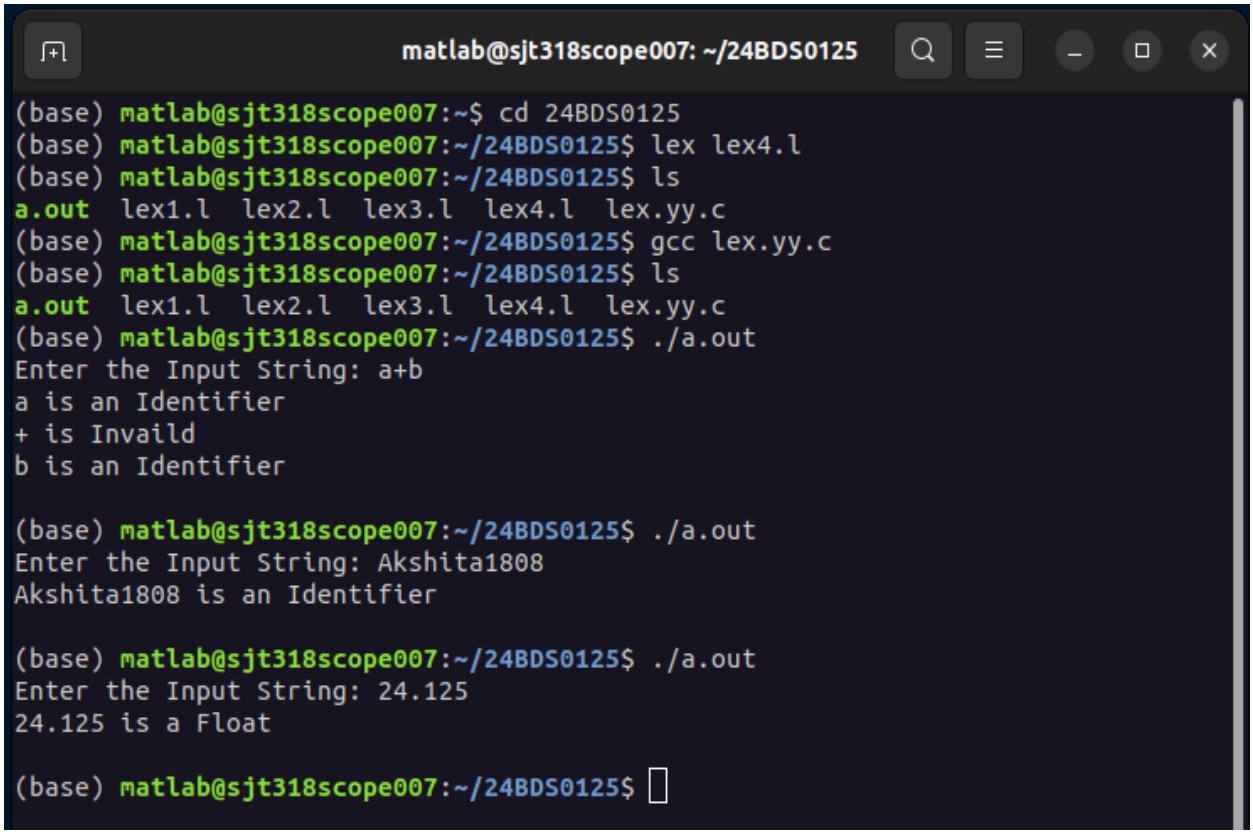
- A valid identifier (starting with a letter or underscore or dollar sign, followed by letters, digits, or underscores)
- A valid number (consisting only of digits)

Program:



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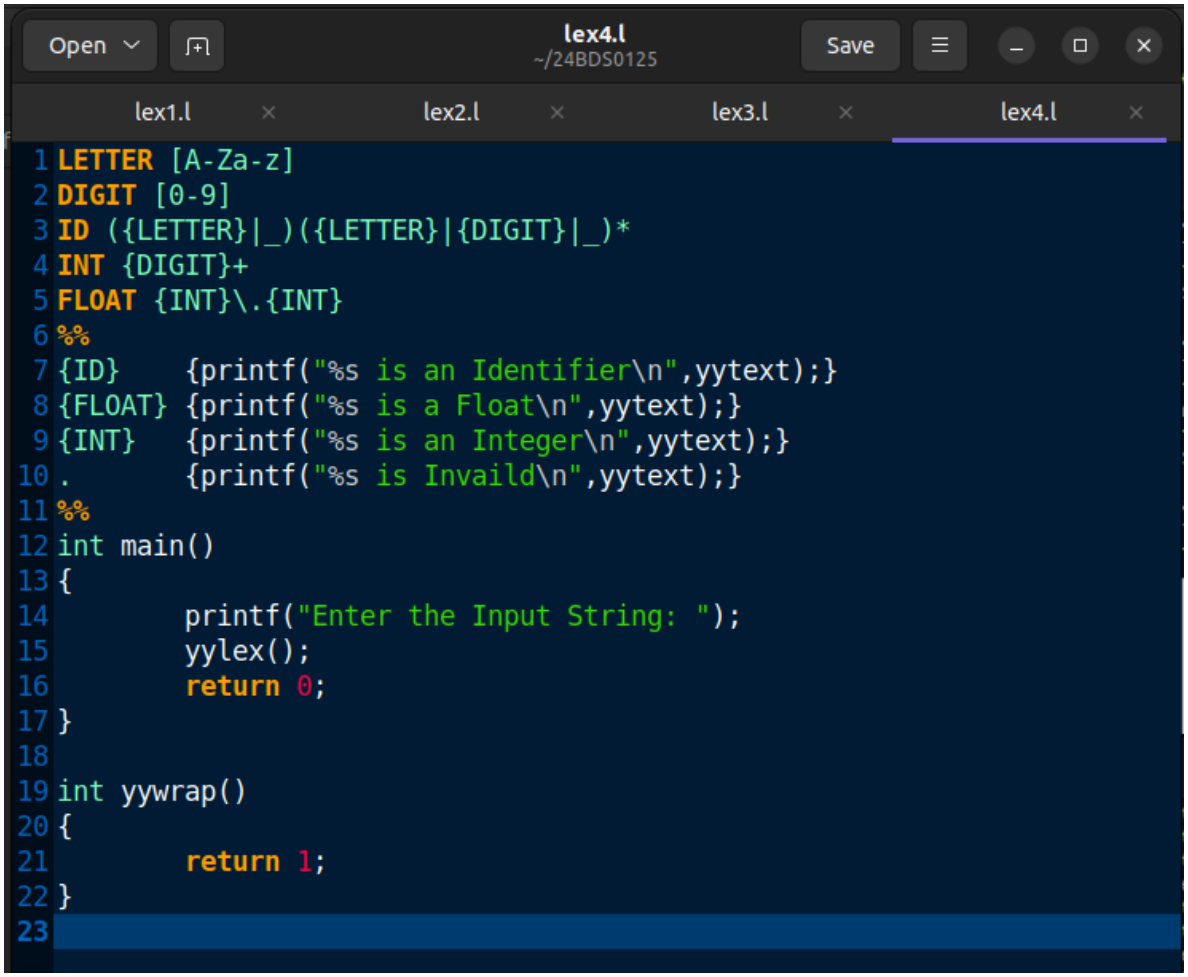
Input/Output Screenshots:



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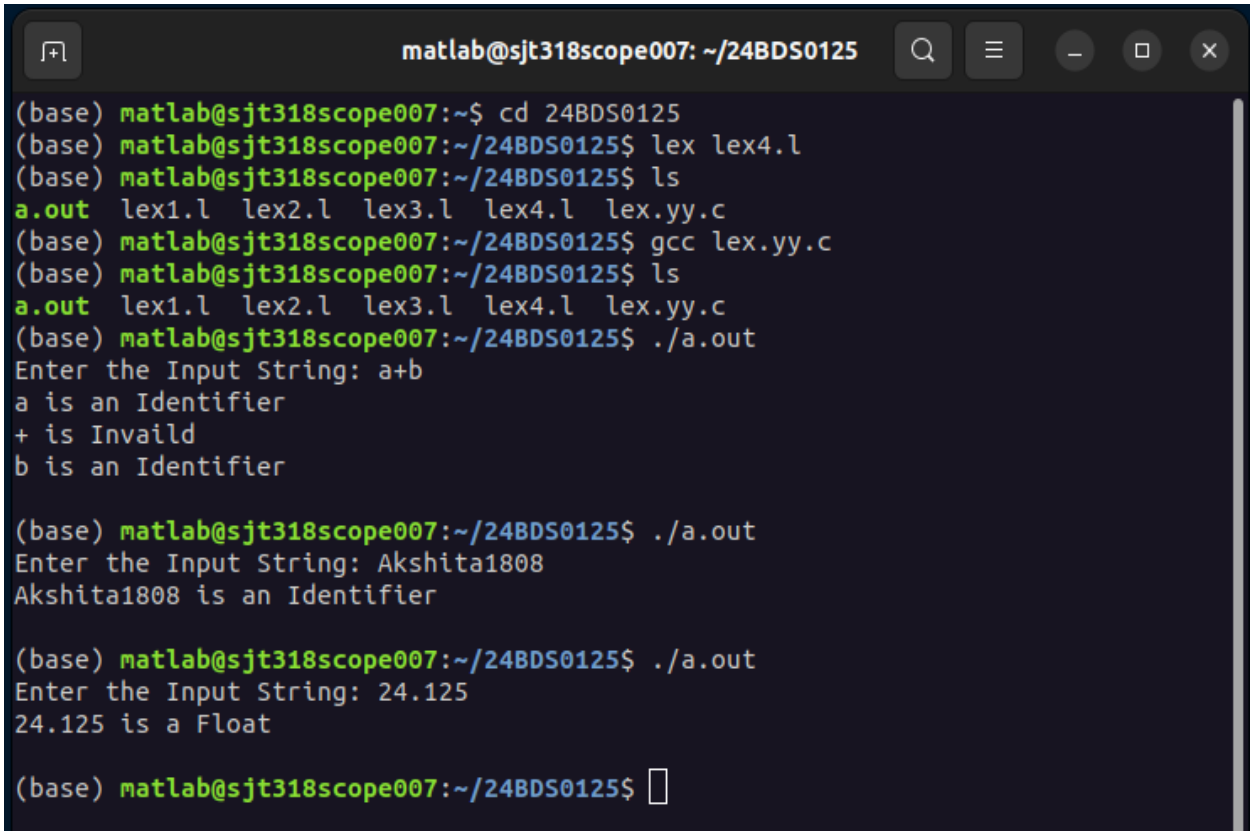
Question 5: A calculator application's backend needs to process user input and determine whether the entered numeric value is an integer or a real (decimal) number before performing calculations. Write a Lex program that reads a numeric input and classifies it as: Integer and Real Number.

Program:



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Input/Output Screenshots:

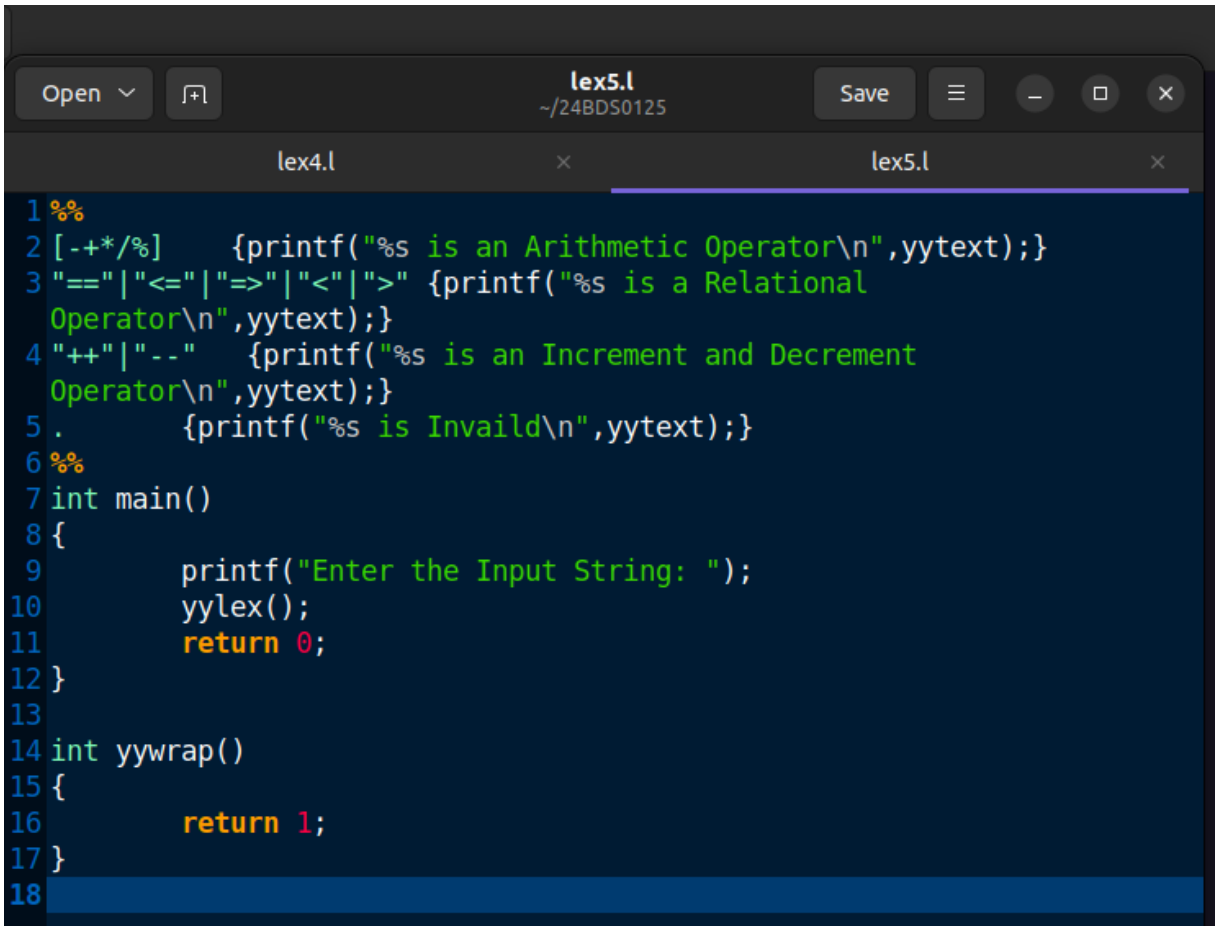


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Question 6: A code highlighter tool is used in an Integrated Development Environment (IDE) to enhance the readability of source code. As part of its functionality, it must detect and classify different types of operators present in the code. Write a Lex program that reads a source code snippet and detects the following categories of operators:

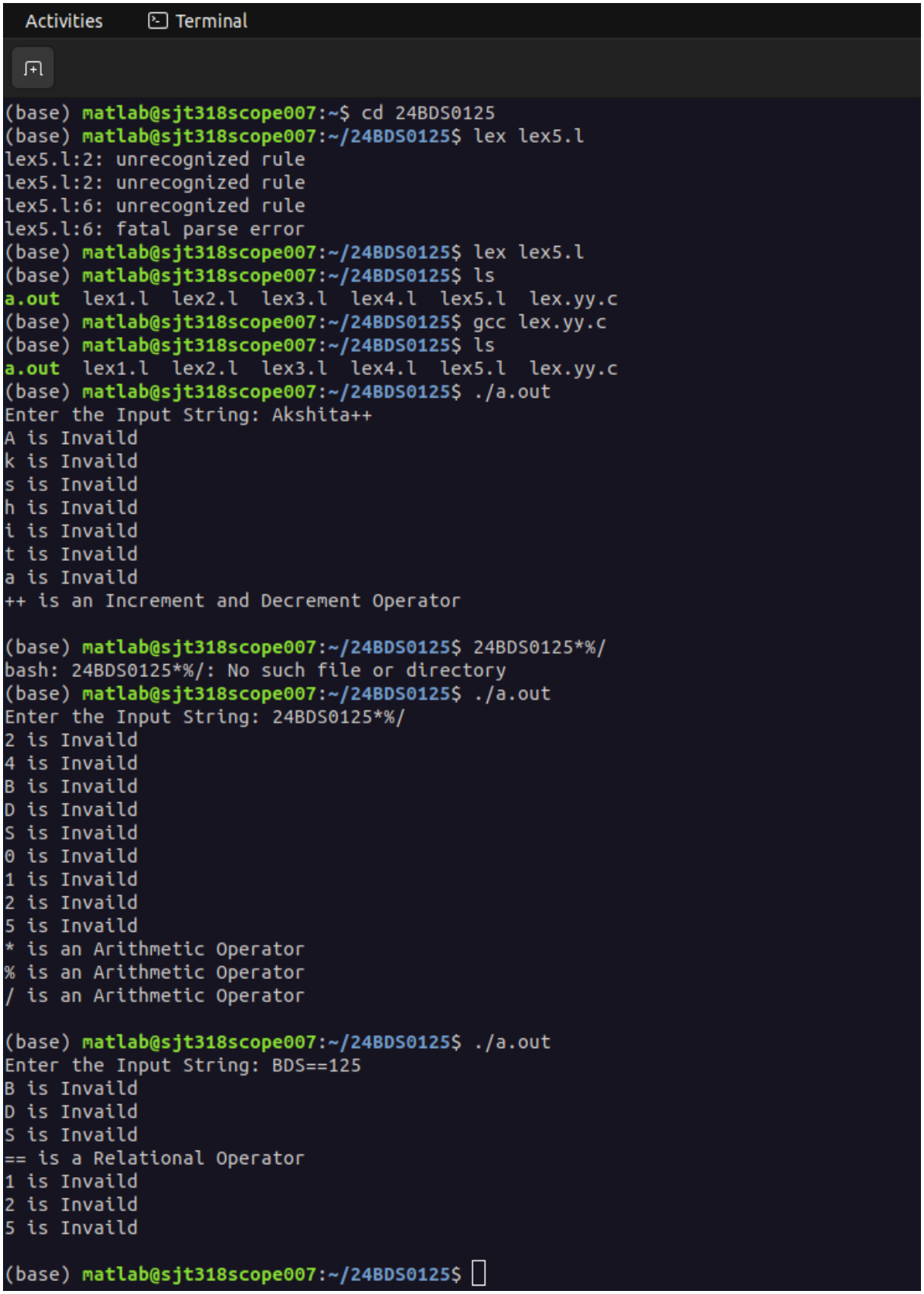
- Arithmetic Operators (+, -, *, /, %)
- Relational Operators (==, !=, >, =, <=)
- Increment and Decrement Operators (++, --)

Program:



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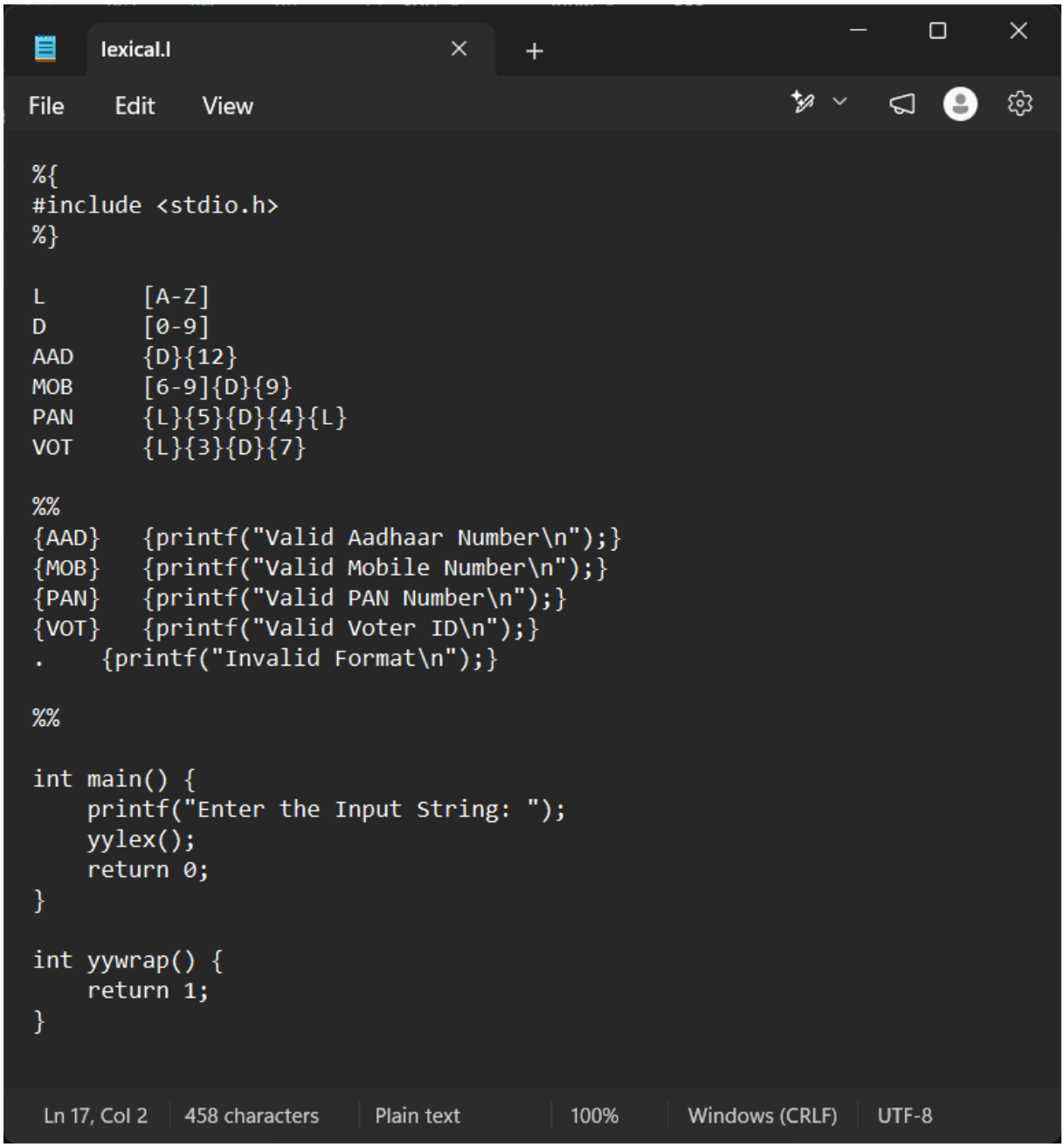
Input/Output Screenshots:



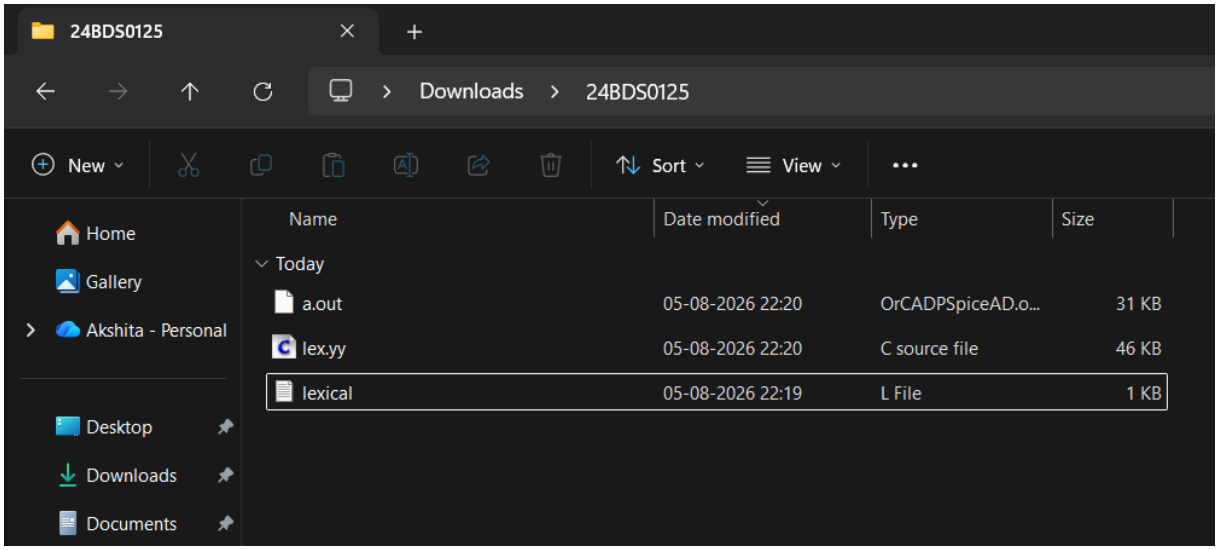
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Question 7: Write a Lex program that validates user input for an online KYC form by checking whether the entered details match the following specified formats:
▪ Aadhar Number: 12-digit numeric format
▪ Mobile Number: 10-digit numeric format starting with digits 6–9
▪ PAN Number: Format ABCDE1234F (5 uppercase letters, 4 digits, 1 uppercase letter)
▪ Voter ID: Format ABC1234567 (3 uppercase letters followed by 7 digits) in their respective valid formats
The program should read each input string and display whether it matches one of the specified valid formats.

Program:



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Input/Output Screenshots:

```
akshita_agarwal@LAPTOP-LP1
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshita Agarwal>wsl
akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal$ cd Downloads
akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads$ cd 24BDS0125
akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ lex lexical
.l
akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ ls
a.out  lex.yy.c  lexical.l
akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ gcc lex.yy.
c
akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ ls
a.out  lex.yy.c  lexical.l
akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ ./a.out
Enter the Input String: 234456789876
Valid Aadhaar Number

akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ ./a.out
Enter the Input String: AkshitaAgarwal
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format
Invalid Format

akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ ./a.out
Enter the Input String: ABCDE1234F
Valid PAN Number

akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$ ./a.out
Enter the Input String: BDS2400125
Valid Voter ID

akshita_agarwal@LAPTOP-LP1C7TFI:/mnt/c/Users/Akshita Agarwal/Downloads/24BDS0125$
```